

◆ 16.3 Energy and Stability Analysis

16.3.1 Lyapunov Functional

$$\mathcal{L}(\Psi, P) = \mathcal{E}(\Psi) + KL(P \parallel P_0)$$

where:

- $\mathcal{E}(\Psi)$: physical energy functional
- $KL(P \parallel P_0)$: relative entropy (Kullback–Leibler divergence)

Thus:

$$\mathcal{L}(\Psi, P) \geq 0$$

and serves as a candidate Lyapunov functional.

16.3.2 Energy Dissipation

$$\begin{aligned} \frac{d\mathcal{L}}{dt} &\leq \\ &- c \|\nabla \mathcal{E}(\Psi)\|^2 \\ &+ C \sigma^2 \end{aligned}$$

This implies:

the system dissipates energy up to stochastic perturbations.

16.3.3 Mean-Square Stability

$$\mathbb{E} \|\Psi_k - \Psi^*\|^2 \leq (1 - c\Delta t)^k \|\Psi_0 - \Psi^*\|^2 + O(\sigma^2)$$

Thus:

the system converges to a bounded neighborhood of equilibrium.

16.3.4 Almost-Sure Stability

If:

$$\sigma(t) \rightarrow 0$$

then:

$$\mathbb{P}(\Psi_k \rightarrow \Psi^*) = 1$$

i.e., convergence holds almost surely.

16.3.5 Invariant Measure

$$\lim_{L \rightarrow \infty} \frac{1}{L} \sum_{i=1}^L \pi(x_i) = 0$$

Thus:

$$\exists \pi(x)$$

a stationary (invariant) probability distribution.

16.3.6 Uniqueness of the Invariant Measure

If:

$$\lim_{L \rightarrow \infty} \frac{1}{L} \sum_{i=1}^L V(x_i) < \infty$$

then:

$$\pi(x) \text{ is unique}$$

16.3.7 Network Stability

$$\lambda_2(L) > 0$$

implies:

the network remains connected and dynamically coherent.

16.3.8 Noise Robustness

$$\mathbb{E}[\|\Psi\|^2] \leq C + O(\sigma^2)$$

Thus:

the system remains bounded under stochastic disturbances.

16.3.9 Numerical Stability

$$\Delta t < \frac{1}{L_{\text{tot}}}$$

ensuring:

absence of numerical divergence.

16.3.10 Link to Dynamics

Since:

$$F \supset -\nabla \mathcal{E}(\Psi)$$

it follows that:

$\mathcal{L}$ is a valid Lyapunov functional for the system dynamics defined in Section 16.2.

Closure Statement (Part III)

$$\boxed{\text{The system is globally stable, dissipative, and robust under stochastic disturbances.}}$$

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